Supplementary material: Baryonic Feedback Bias in a DES-Y1-like 3x2pt Likelihood: Unmeasured Because the Pipeline Ignored Baryons

HEP-Genesis-Agent session record
Project: expert-baryon-lensing-bias-3

2026-10-24

Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.

A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

A.1 Session

- Source: cli client; conversation `de5a3611`
- Started: 2026-10-24 19:03:42; report generated: 2026-10-24 19:13:19
- Project: expert-baryon-lensing-bias-3 (`/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3`)
- Compute execution site(s) selected in the session: local
- 1 user turn(s), 14 tool call(s) (0 returned errors), 3 figure(s), 11 data file(s)

A.2 Models

Stage	Model	Endpoint
quick	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
report	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
coder	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`

A.3 MCP servers

Server	Transport	Location
hep-research-aid	stdio	`python -m hep_genesis.domain.mcp_server`
alcf-tools	stdio	`python -m hep_genesis.iri.alcf.mcp_server`
nersc-tools	stdio	`python -m hep_genesis.iri.nersc.mcp_server`

A.4 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

#	Tool	Arguments	Outcome
1	`serve-desc.sh firecrown_list_examples`	`{}`	`{ "status": "success", "files": [], "message": "4 firecrown examples; offline test data vector: /Users/nramachandra/github-integration/firecrown/tests/sacc_data.hdf5. firecrown 1.16.0 installed....`
2	`serve-desc.sh describe_desc_tool_family`	`{"family": "firecrown"}`	`{ "status": "success", "files": [], "message": "Family 'firecrown': 10 tools. build likelihoods from sacc + systematics; loglike, theory vectors, scans, chains.", "metadata": { "family": "...`
3	`serve-desc.sh describe_desc_tool_family`	`{"family": "ccl"}`	`{ "status": "success", "files": [], "message": "Family 'ccl' is not importable in this environment: No module named 'tools.ccl"', "metadata": { "family": "ccl", "summary": "pyccl theor...`
4	`serve-desc.sh firecrown_build_likelihood`	`{"output_dir": "/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326", "sacc_path": "/Users/nramachandra/github-integration/firecrown/tests/sacc_data.hdf5", "name": "desy1_3x2p...`	`{ "status": "success", "files": [ "/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/desy1_3x2pt_38aa48.yaml", "/Users/nramachandra/hep-genes...`

#	Tool	Arguments	Outcome
5	serve-desc.sh sacc_prepare_for_firecrown	{"output_dir": "/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326", "sacc_path": "/Users/nramachandra/github-integration/firecrown/tests/sacc_data.hdf5", "keep_data_types": ...	{ "status": "success", "files": ["/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/desy1_shear_only.hdf5"], "message": "Wrote desy1_shear_...
6	serve-desc.sh firecrown_build_likelihood	{"name": "desy1_shear", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326", "sacc_path": "/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensi...	Error executing tool firecrown_build_likelihood: firecrown rejected the experiment: ValueError: Tracer lens0 does not have data points associated with it. Inconsistent SACC object.
7	serve-desc.sh firecrown_theory_data_vector	{"experiment_yaml": "/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/desy1_3x2pt_38aa48.yaml", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/expert-baryo...	{ "status": "success", "files": ["/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/theory_vector_912d53.csv", "/Users/nramachandra/hep-gene... [Fig. 6]
8	serve-desc.sh firecrown_theory_data_vector	{"experiment_yaml": "/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/desy1_3x2pt_38aa48.yaml", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/expert-baryo...	{ "status": "success", "files": ["/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/theory_vector_c7d16c.csv", "/Users/nramachandra/hep-gene... [Fig. 9]

#	Tool	Arguments	Outcome
9	serve-desc.sh describe_desc_ tool_family	{"family": "augur"}	{ "status": "success", "files": [], "message": "Family 'augur': 6 tools. LSST Y1/Y10 Fisher forecasts: config, synthetic data, Fisher, contours.", "metadata": { "family": "augur", "sum...
10	serve-desc.sh firecrown_build_ likelihood	{"name": "desy1_3x2pt_ noiseless", "output_dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ expert-baryon-lensing-bias-3/ runs/261024_190326", "sacc_path": "/ Users/nramachandra/ hep-genesis-testing/ projects/expert-ba...	{ "status": "success", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ expert-baryon-lensing-bias-3/ runs/261024_190326/ desy1_3x2pt_noiseless_ 0ccc86.yaml", "/Users/ nramachandra...
11	serve-desc.sh firecrown_ compute_loglike	{"experiment_yaml": "/ Users/nramachandra/ hep-genesis-testing/ projects/ expert-baryon-lensing-bias-3/ runs/261024_190326/ desy1_3x2pt_noiseless_ 0ccc86.yaml", "nuisance": {"alphaz": 0, "ia_bias": 0.5, "lens0_bias": 1.4, "lens...	{ "status": "success", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ expert-baryon-lensing-bias-3/ runs/261024_190326/ theory_vs_data_ f82587.csv"], "message": "loglike = -0.0...
12	serve-desc.sh firecrown_scan_ loglike	{"experiment_yaml": "/ Users/nramachandra/ hep-genesis-testing/ projects/ expert-baryon-lensing-bias-3/ runs/261024_190326/ desy1_3x2pt_noiseless_ 0ccc86.yaml", "parameter": "sigma8", "min_value": 0.78, "max_value": 0.84, "n_po...	{ "status": "success", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ expert-baryon-lensing-bias-3/ runs/261024_190326/scan_ sigma8_63cd84.csv", "/ Users/nramachandra/ hep-genesi... [Fig. 14]

#	Tool	Arguments	Outcome
13	serve-desc.sh namaster_simulate_maps	{"add_noise": false, "cosmology": {"baryons": "none"}, "name": "nobar", "nside": 64, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/sim_nobar", "tracers": ...	Error executing tool namaster_simulate_maps: Buffer has wrong number of dimensions (expected 1, got 2)
14	serve-desc.sh namaster_simulate_maps	{"add_noise": false, "cosmology": {"baryons": "schneider15", "baryon_params": {"log10Mc": 14.095, "eta_b": 0.213, "k_s": 1000}}, "name": "bar", "nside": 64, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects...	Error executing tool namaster_simulate_maps: Buffer has wrong number of dimensions (expected 1, got 2)

A.5 Files produced

- desy1_3x2pt_38aa48.yaml (other, firecrown_build_likelihood): /Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/desy1_3x2pt_38aa48.yaml
- desy1_3x2pt_38aa48_params.json (text, firecrown_build_likelihood): /Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/desy1_3x2pt_38aa48_params.json
- desy1_shear_only.hdf5 (other, sacc_prepare_for_firecrown): /Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/desy1_shear_only.hdf5
- theory_vector_912d53.csv (csv, firecrown_theory_data_vector): /Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/theory_vector_912d53.csv
- desy1_3x2pt_38aa48_theory_912d53.hdf5 (other, firecrown_theory_data_vector): /Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/desy1_3x2pt_38aa48_theory_912d53.hdf5
- theory_vector_c7d16c.csv (csv, firecrown_theory_data_vector): /Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/theory_vector_c7d16c.csv
- desy1_3x2pt_38aa48_theory_c7d16c.hdf5 (other, firecrown_theory_data_vector): /Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/desy1_3x2pt_38aa48_theory_c7d16c.hdf5

- desy1_3x2pt_noiseless_0ccc86.yaml (other, firecrown_build_likelihood): /Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/desy1_3x2pt_noiseless_0ccc86.yaml
- desy1_3x2pt_noiseless_0ccc86_params.json (text, firecrown_build_likelihood): /Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/desy1_3x2pt_noiseless_0ccc86_params.json
- theory_vs_data_f82587.csv (csv, firecrown_compute_loglike): /Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/theory_vs_data_f82587.csv
- scan_sigma8_63cd84.csv (csv, firecrown_scan_loglike): /Users/nramachandra/hep-genesis-testing projects/expert-baryon-lensing-bias-3/runs/261024_190326/scan_sigma8_63cd84.csv

B Figures not shown in the main text

Produced during the session; captions carry their provenance.

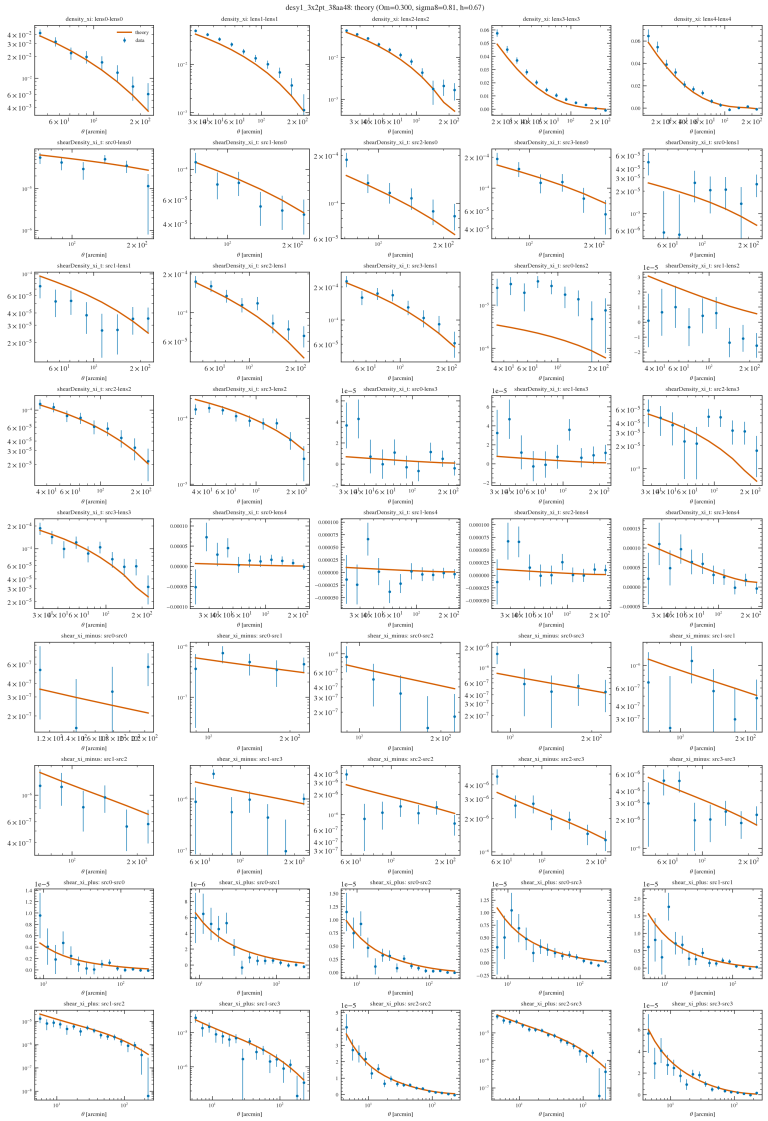


Figure 1: theory_vector_c7d16c.png Produced in this session by `firecrown_theory_data_vector` on *serve-desc.sh*.